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Dual tubing locating adaptor for a tubing string

Abstract

A dual tubing locating adapter ("DLTA") for use with a mule shoe. The DLTA may include a housing, a primary tubing string, a lateral tubing string, and a retention mechanism. The primary tubing string may be positioned at least partially within the DLTA. The lateral tubing string may be positioned at least partially within the DLTA and retained within the DLTA via a shear mechanism. The retention mechanism may be operable to retain the lateral tubing string within the DLTA as the DLTA is landed on the mule shoe.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage Entry of International Patent Application No. PCT/US2022/052636, filed Dec. 13, 2022, which claims the benefit of U.S. Provisional Application No. 63/293,222 entitled "Dual Tubing Locating Adaptor with Lateral Bore and Main Bore Tubing Strings," filed Dec. 23, 2021, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND
(1) A wellbore may be drilled into the ground to explore and produce a hydrocarbon reservoir therein. This wellbore may be referred to as the main or primary wellbore. To further explore and/or increase production from the reservoir, one or more lateral wellbores may be drilled which branch from the main wellbore. Such drilling extends the reach of the well into laterally displaced portions of the reservoir.
(2) During downhole operations, it may be necessary to separately and selectively enter the main and lateral wellbores with a wellbore tubular or tubulars. The wellbore tubulars, or tubing strings, can be used to establish flow or access paths in the multiple wellbores. For example, production strings can be guided to the main and lateral wellbores, and sealed, to provide fluid flow paths from the multiple wellbores into the primary well extending to the surface.

SUMMARY
(3) A dual tubing locating adapter ("DLTA") for use with a mule shoe according to one or more embodiments of the present disclosure includes a housing, a primary tubing string, a lateral tubing string, and a retention mechanism. The primary tubing string is positioned at least partially within the DLTA. The lateral tubing string is positioned at least partially within the DLTA and retained within the DLTA via a shear mechanism. The retention mechanism is operable to retain the lateral tubing string within the DLTA as the DLTA is landed on the mule shoe.
(4) A production tubing system for use within a wellbore according to one or more embodiments of the present disclosure includes a mule shoe disposable within the wellbore and a DLTA. The DLTA includes a housing, a primary tubing string, a lateral tubing string, and a retention mechanism. The primary tubing string is positioned at least partially within the DLTA. The lateral tubing string is positioned at least partially within the DLTA and retained within the DLTA via a shear mechanism. The retention mechanism is operable to retain the lateral tubing string within the DLTA as the DLTA is landed on the mule shoe.
(5) A method of completing a wellbore according to one or more embodiments of the present disclosure includes positioning a mule shoe within a wellbore. The method further includes running a DLTA comprising a primary tubing string and a lateral tubing string into the wellbore. The method also includes landing the DLTA on the mule shoe while retaining the lateral tubing string at least partially within the DLTA via a retention mechanism and a shear mechanism
(6) However, many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Certain embodiments of the disclosure will hereafter be described with reference to the accompanying drawings, wherein like reference numerals denote like elements. It should be understood, however, that the accompanying figures illustrate the various implementations described herein and are not meant to limit the scope of various described technologies. The drawings are as follows:
(2) FIG. **1** is a schematic view of a system for milling and drilling a lateral wellbore from a primary wellbore according to one or more embodiments of the present disclosure;
(3) FIG. **2** is a schematic view of the finished junction between the lateral wellbore and the primary wellbore including downhole operations equipment according to one or more embodiments of the present disclosure;
(4) FIG. **3** is a schematic view of an embodiment of a production tubing assembly according to one or more embodiments of the present disclosure;
(5) FIG. **4** is an isometric view of a dual tubing locating adapter ("DLTA") according to one or more embodiments of the present disclosure;
(6) FIG. **5** is a cross-sectional view of the DLTA of FIG. **4** along line A-A;
(7) FIG. **6** is a cross-sectional view of the DLTA of FIG. **4** along line B-B;
(8) FIG. **7** is an enlarged, partial cross-sectional view of the DLTA of FIG. **4**;
(9) FIG. **8** is a side view of the DLTA of FIG. **4** in the process of landing on a mule shoe;
(10) FIG. **9** is an isometric view of the DLTA of FIG. **4** in a landed position on the mule shoe.
DETAILED DESCRIPTION
(11) In the following description, numerous details are set forth to provide an understanding of some embodiments of the present disclosure. However, it will be understood by those of ordinary skill in the art that that embodiments of the present disclosure may be practiced without these

details and that numerous variations or modifications from the described embodiments may be possible.

(12) In the specification and appended claims: the terms “connect,” “connection,” “connected,” “in connection with,” “connecting,” “couple,” “coupled,” “coupled with,” and “coupling” are used to mean “in direct connection with” or “in connection with via another element.” As used herein, the terms “up” and “down,” “upper” and “lower,” “upwardly” and “downwardly,” “upstream” and “downstream,” “uphole” and “downhole,” “above” and “below,” and other like terms indicating relative positions above or below a given point or element are used in this description to more clearly describe some embodiments of the disclosure.

(13) Referring initially to FIG. **1**, FIG. **1** is a system **100** for milling and drilling a lateral wellbore **102** from a primary wellbore **104**. A primary or main wellbore **104** is drilled in a conventional manner and may include operational equipment **106**, such as a whipstock and anchor system, or a fracturing and/or production system **108**. A diverter or whipstock **110** is used to guide a milling and/or drilling assembly **112** laterally relative to the primary wellbore **104** for creating a lateral or secondary wellbore **102** having a junction **114** with the primary wellbore **104**. After the junction and lateral wellbore are complete, well treatment, completion or production equipment **108** may remain in the primary wellbore **104** along with an orientator or locator **200** for receiving additional downhole tools, as shown in FIG. **2**.

(14) Referring now to FIG. **3**, FIG. **3** is a production tubing system **300** according to one or more embodiments of the present disclosure. The production tubing assembly **300** is adapted for providing a pressure seal, which isolates the lateral wellbore **102** from the main wellbore **104** and vice versa, to the two (or multiple) bores for production access. In one or more embodiments, the production tubing system **300** aligns on a mule shoe **302** downhole and latches to the top of the junction **114**. When latched, a dual tubing locating adapter (“DLTA”) **304** with dual strings **306**, **308** advances into the junction **114**.

(15) Once the DLTA **304** is in place in the junction **114**, the DLTA selectively guides the strings **306**, **308** into the main and lateral bores **104**. The primary bore string **306** lands in a polished bore receptacle **310** of the primary bore production equipment **108** and the lateral bore string **308** lands in a polished bore receptacle **312** of the secondary bore production equipment **314**. For purposes of simplicity and clarity, the strings **306**, **308** and the equipment **108**, **314** will be referred to as production strings and equipment, though other tubular members and downhole equipment are contemplated. The positioned production tubing assembly **300** and production strings **306**, **308** will effect a seal in the bores of the production equipment **108**, **314** in the main and lateral wellbores **104**, **102** to complete the well. A packer assembly **316** and other downhole equipment may also be provided in the wellbores **104**, **102**.

(16) In some embodiments, a diverter **318** is disposed at the top of the production tubing assembly **300** that selectively allows access to either bore for future intervention work needed downhole. The diverter **318** may stay in place and can be rotated by means of multi-cycle “J” grooves to allow access to the desired bore. A packer **320**, with a seal bore receptacle, is set at the top of the production tubing assembly **300** to lock the assembly in place. If another junction is created in the main wellbore **104** above the original junction **114**, a packer is provided to seal access to the lower junction **114**, making the production tubing assembly **300** stackable.

(17) Referring now to FIG. **4**, FIG. **4** is an isometric view a DLTA **404** according to one or more embodiments of the present disclosure. In one or more embodiments, the DLTA **404** includes a housing **400** made of an upper portion **402** and a lower portion **406** joined via pins, screws, or similar connectors. Additionally, a mule shoe ramp **408** is formed on the exterior of the housing **400** to align the DLTA on a mule shoe, as described in more detail below. A primary tubing string **410** and the lateral tubing string **412** extend through the housing **400**. The DLTA **404** also includes locking collet fingers **414** to couple the DLTA **404** to a mule shoe once the DLTA **404** is landed and a retention mechanism **416** to prevent premature release of the lateral tubing string **412**, as

described in more detail below.
(18) Turning now to FIG. **5**, FIG. **5** is a cross-sectional view of the DLTA **404** of FIG. **4** along line A-A. In one or more embodiments, the housing **400** is positioned around the primary tubing string **410** and the lateral tubing string **412**. The lateral tubing string **412** includes two protrusions **502** extending at least a portion of the length of the lateral tubing string **412** to prevent relative rotation between the lateral tubing string and the DLTA **404**. In other embodiments, the lateral tubing string **412** may include one, three, or more protrusions **502** to prevent relative motion between the lateral tubing string **412** and the DLTA **404**. The lateral tubing string is retained within the housing via a shear mechanism **500**, such as shear screws extending through the housing **400**. Once the DLTA **404** is landed on the mule shoe, as described in more detail below, the shear mechanism is sheared to allow the lateral tubing string to extend into the lateral wellbore.
(19) Turning now to FIG. **6**, FIG. **6** is a cross-sectional view of the DLTA **404** of FIG. **4** along line B-B. As shown in FIG. **6**, the retention mechanism **416** includes a lug **600** retained partially within the housing **400** via retaining pins **602** or a similar mechanism and the lug **600** is biased radially outward via a biasing mechanism **604**, such as a spring. As shown more clearly in FIGS. **7** and **8**, the lug **600** extends beyond the housing **400** such that when the DLTA begins to land on the mule shoe **800**, the lug is shifted via the mule shoe **800** and engages with a retention profile **700** formed on the lateral tubing string **412**. This allows the lug **600** to retain the lateral tubing string **412** within the DLTA **404** as the DLTA **404** is landed.
(20) Once the DLTA **404** is landed and aligned via the mule shoe **800** and the mule shoe ramp **408**, the lug **600** is exposed, as shown in FIG. **9**. Once the lug is exposed, the biasing mechanism **604** extends the lug **600** away from the lateral tubing string **412**, allowing the lateral tubing string **412** to extend into the lateral wellbore, as described above. Additionally, the collet fingers **414** of the DLTA extend through slots **900** formed in the mule shoe **800** to couple the DLTA **404** to the mule shoe **800**. In one or more embodiments, once the DLTA is landed on the mule shoe, the primary tubing string **410** extends into the primary wellbore, as described above.
(21) As used herein, a range that includes the term between is intended to include the upper and lower limits of the range; e.g., between 50 and 150 includes both **50** and **150**. Additionally, the term “approximately” includes all values within 5% of the target value; e.g., approximately 100 includes all values from 95 to 105, including 95 and 105. Further, approximately between includes all values within 5% of the target value for both the upper and lower limits; e.g., approximately between 50 and 150 includes all values from 47.5 to 157.5, including 47.5 and 157.5.
(22) Although a few embodiments of the disclosure have been described in detail above, those of ordinary skill in the art will readily appreciate that many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Claims

1. A dual tubing locating adapter (“DLTA”) for use with a mule shoe, the DLTA comprising: a housing; a primary tubing string positioned at least partially within the DLTA; a lateral tubing string positioned at least partially within the DLTA and retained within the DLTA via a shear mechanism; and a retention mechanism operable to retain the lateral tubing string within the DLTA as the DLTA is landed on the mule shoe, wherein the retention mechanism comprises: a lug retained within the housing and shiftable to engage with a retention profile formed on the lateral tubing string to retain the lateral tubing string within the DLTA; and a biasing mechanism to bias the lug radially outward, wherein the lug is configured to be compressed via the mule shoe when landing the DLTA on the mule shoe such that the lug engages with the retention profile formed on the lateral tubing string to retain the lateral tubing string within the DLTA.
2. The DLTA of claim 1, wherein the housing further comprises a mule shoe ramp formed on an

exterior of the housing to align the DLTA with the mule shoe.
3. The DLTA of claim 1, wherein the biasing mechanism comprises a spring.
4. The DLTA of claim 1, further comprising collet fingers positioned to couple the DLTA with the mule shoe once the DLTA is landed on the mule shoe.
5. The DLTA of claim 1, wherein the shear mechanism comprises shear screws extending through the housing to retain the lateral tubing string within the DLTA.
6. The DLTA of claim 1, wherein the housing comprises an upper portion coupled to a lower portion.
7. A production tubing system for use within a wellbore; the production tubing system comprising: a mule shoe disposable within the wellbore; and a dual tubing locating adapter ("DLTA") disposable within the wellbore and comprising: a housing; a primary tubing string positioned at least partially within the DLTA; a lateral tubing string positioned at least partially within the DLTA and retained within the DLTA via a shear mechanism; and a retention mechanism operable to retain the lateral tubing string within the DLTA as the DLTA is landed on the mule shoe, wherein the retention mechanism comprises: a lug retained within the housing and shiftable to engage with a retention profile formed on the lateral tubing string to retain the lateral tubing string within the DLTA; and a biasing mechanism to bias the lug radially outward, wherein the lug is configured to be compressed via the mule shoe when landing the DLTA on the mule shoe such that the lug engages with the retention profile formed on the lateral tubing string to retain the lateral tubing string within the DLTA.
8. The production tubing system of claim 7, wherein the housing further comprises a mule shoe ramp formed on an exterior of the housing to align the DLTA with the mule shoe.
9. The production tubing system of claim 7, wherein the biasing mechanism comprises a spring.
10. The production tubing system of claim 7, wherein the DLTA further comprises collet fingers positioned to couple the DLTA with the mule shoe once the DLTA is landed on the mule shoe.
11. The production tubing system of claim 7, wherein the shear mechanism comprises shear screws extending through the housing to retain the lateral tubing string within the DLTA.
12. The production tubing system of claim 7, wherein the housing comprises an upper portion coupled to a lower portion.
13. A method of completing a well, the method comprising: positioning a mule shoe within a wellbore; running a dual tubing locating adapter ("DLTA") comprising a primary tubing string and a lateral tubing string into the wellbore; landing the DLTA on the mule shoe while retaining the lateral tubing string at least partially within the DLTA via a retention mechanism and a shear mechanism; and biasing a lug of the retention mechanism radially outward via a biasing mechanism, wherein landing the DLTA on the mule shoe further comprises compressing the lug via the mule shoe such that the lug engages with a retention profile formed on the lateral tubing string to retain the lateral tubing string within the DLTA.
14. The method of claim 13, further decompressing the lug of the retention mechanism radially outward to allow the lateral tubing string to be positioned within a lateral wellbore of the well.